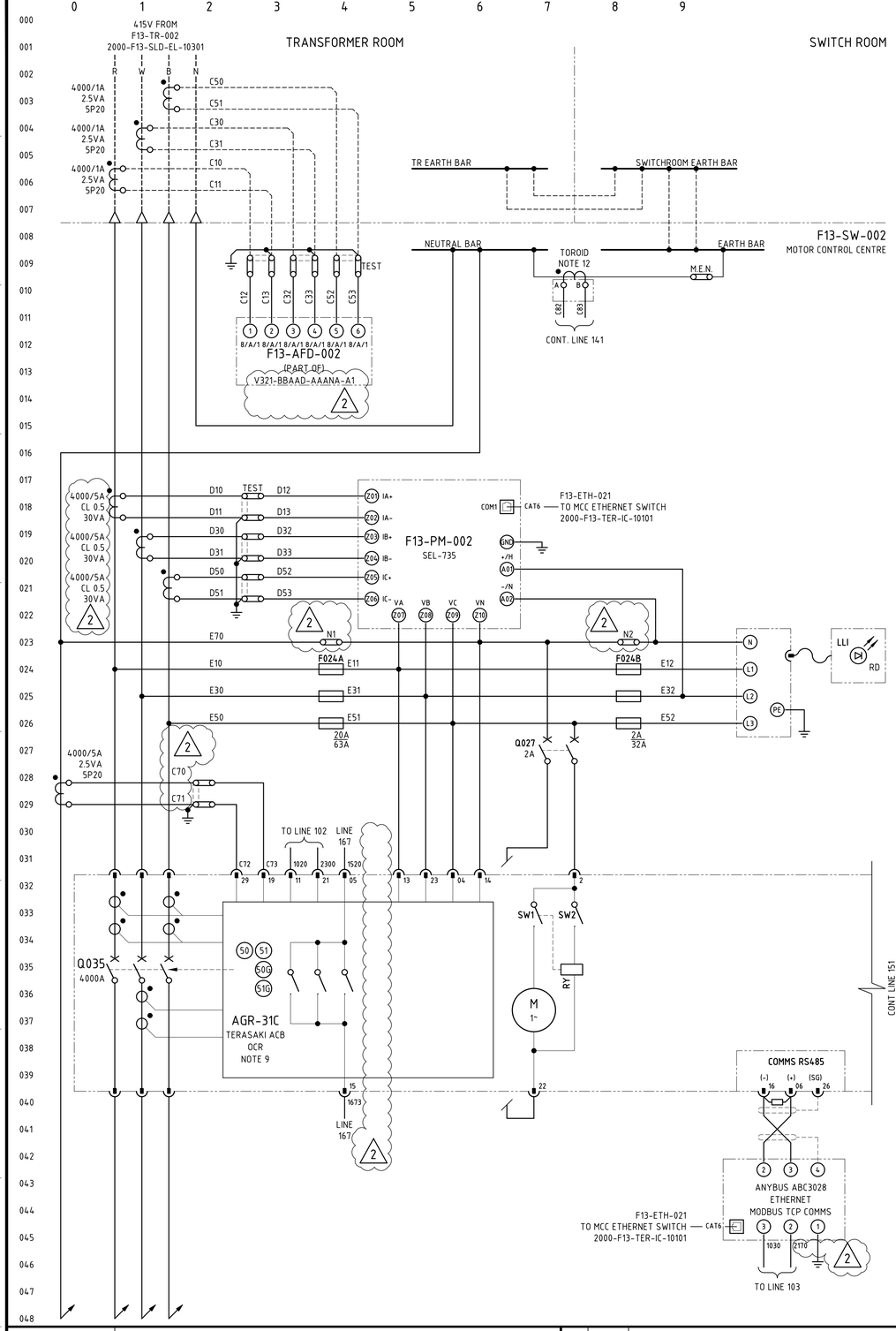
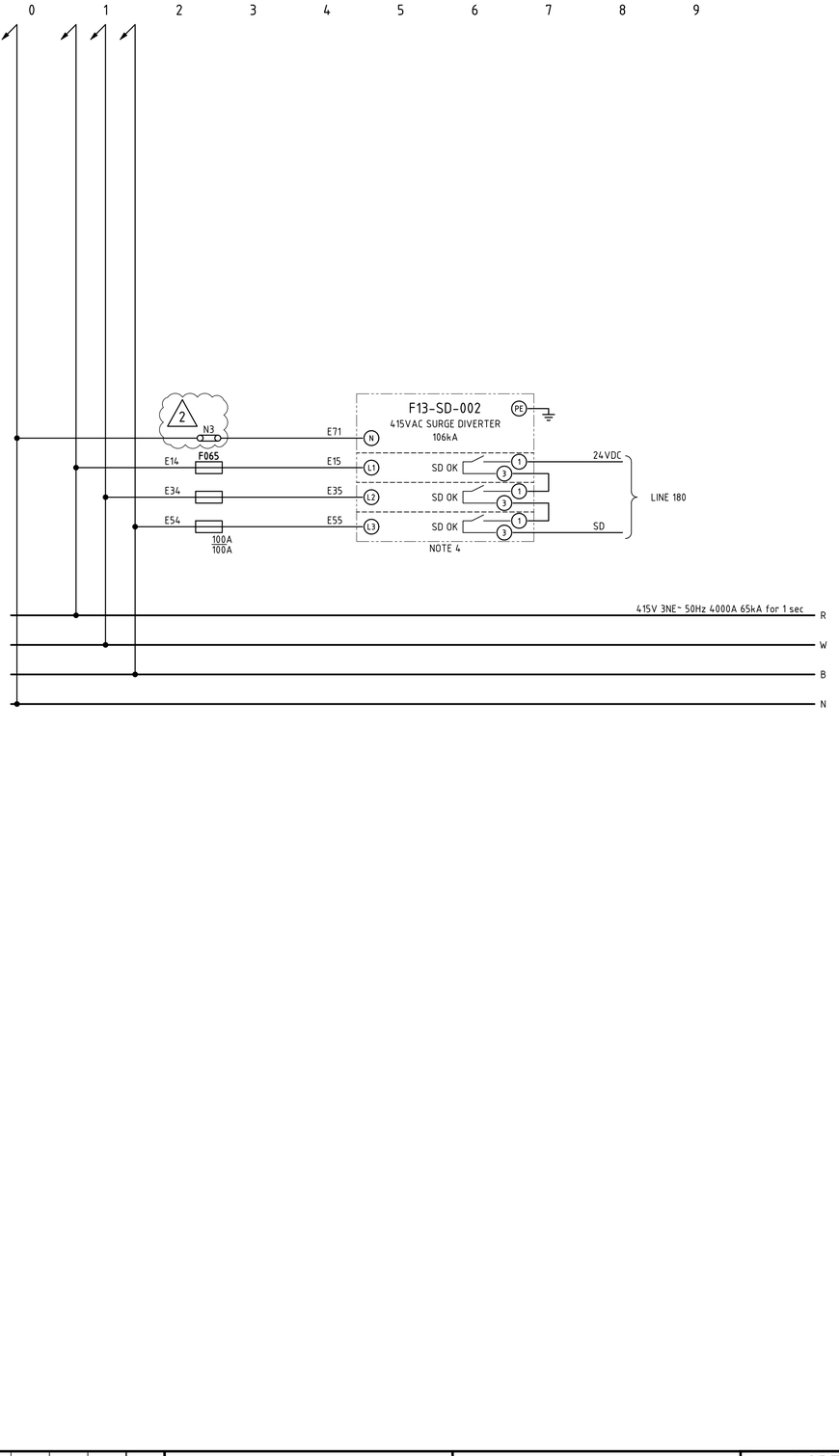



## NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT FOR INCOMER ZONE. 415VAC MCC INCOMER TO BE TRIPPED FOR BUSBAR, OUTGOING MODULE AND OUTGOING CABLE ZONE ARC FAULTS.
3. ALL CIRCUIT BREAKER STATUS VIA DRY CONTACT TO DCS CUBICLE.
4. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
5. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
6. ALL TERMINATIONS ABOVE 50VAC SHALL BE TO IP2x OR ABOVE, IF NOT, TO HAVE A SEPARATE PERSPEX SHROUD FITTED AND SHALL BE LABELLED "DANGER xxxVAC".
7. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F13SW002-0600, UNLESS OTHERWISE NOTED.
8. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
9. SPLIT CIRCUIT BETWEEN LATCH RELEASE COIL (LRC), MODBUS OPEN/ CLOSE, RACKING MOTOR AND UNDER VOLTAGE TRIP OPTION TO BE SPECIFIED WHEN ORDERING.
10. REMOTE CIRCUIT BREAKER OPEN/ CLOSE AND CIRCUIT BREAKER RACKING FROM DCS IS DISABLED: THIS FEATURE MAY BE ENABLED BY NSR IN THE FUTURE.
11. ALL CIRCUIT BREAKER PROTECTION, ARC FLASH DETECTION (AFD) AND POWER METER STATUS VIA MODBUS TCP.FOR COMMUNICATION BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10011.
12. EARTH LEAKAGE RELAY USER INTERFACE TO BE MOUNTED ON MCC MODULE DOOR. RELAY ENERGISED (NORMALLY CLOSED CONTACT OPEN) IN NORMAL OPERATION.
    NHP TDGC2
    IN:170A IΔN MIN:0.05A TOROID TO SUIT CABLE DIAMETER.
13. PHASE CT's FOR ARC FLASH DETECTION PROTECTION AND CURRENT SUPERVISION LOCATED IN UPSTREAM DRY TRANSFORMER ENCLOSURE TO AVOID AND MCC 'BLIND SPOT' FOR CURRENT SUPERVISED LIGHT SENSOR BASED SCHEME.

## LEGEND

- 00/01/00 — DCS DIGITAL INPUT
- 00/02/00 — DCS DIGITAL OUTPUT
- MCC TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL
- INDICATION LAMP, cc = COLOUR

| DRG No. | REFERENCES |
|---|---|
| 2000-F13-SLD-EL-10301 | F13-SW-002 415V MCC SINGLE LINE DIAGRAM SHEET 1 |
| 2000-F00-STD-EL-10001 | GENERAL NOTES AND STANDARD DRAWINGS INDEX |

| REV | DATE | DESCRIPTION | DRAW | CHECK | DESIGN | TECH APP | PROJ APP |
|---|---|---|---|---|---|---|---|
| 2 | 01AUG25 | FAT REDLINES INCORPORATED | | | | | |
| 1 | 11APR25 | GENERATOR INCOMER ELR SHUNT TRIP ADDED | | | | | |
| 0 | 07AUG24 | APPROVED FOR CONSTRUCTION | | | | | |

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION - F13-SW-002
F13-SW-002 INCOMER
SCHEMATIC DIAGRAM SHEET 1

| SHEET | SCALE | DRG No. | REVISION |
|---|---|---|---|
| A1 | NONE | 2000-F13-SCM-EL-10301 | 2 |

D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F13 - MILLING & CLASSIFICATION\2000-F13-SCM-EL-10301.DWG 27AUG25 12:14PM HOTSEAT04